

Statistics Tests Reference Guide

A Comprehensive Overview with Real-Life Examples

Overview: This guide covers 14 essential statistical tests used in data analysis, organized by data type and application. Each test includes practical, real-world examples to illustrate its usage.



Test Categories

Continuous Data Tests (1-7)

1. One-sample z-test	2. One-sample t-test
3. Two-sample t-test	4. Paired t-test
5. One-way ANOVA	6. Two-way ANOVA
7. Variance tests (1 & 2 sample)	

Categorical & Count Data Tests (8-14)

8. One-sample proportion	9. Two-sample proportion
10. Chi-square goodness-of-fit	11. Chi-square test of association
12. One-sample Poisson	13. Two-sample Poisson
14. Correlation tests	



Detailed Test Descriptions with Examples

STATISTICAL TEST	REAL-LIFE EXAMPLE
One-Sample Z-Test	A cereal manufacturer states that each box contains 500 g of product. An auditor samples 120 boxes (population SD known = 8 g) to test whether the mean weight actually equals 500 g .
One-Sample T-Test	A university claims the average GPA of graduating students is 3.2 . A professor samples 30 students (population SD unknown) to test if the mean differs from 3.2.
Two-Sample T-Test (Independent)	A hospital compares average blood pressure reduction between patients using medication A vs medication B . Two independent groups, continuous outcome.
Paired T-Test	A company measures employee productivity before and after switching to a new software system. The same employees are tested twice → paired differences.
One-Way ANOVA	A retail chain compares monthly sales performance of stores using three different store layouts (layout X, Y, Z) to check if mean sales differ across the layouts.
Two-Way ANOVA	A school analyses student test scores based on two factors: teaching method (digital/traditional) and grade level (8th/9th) to evaluate main effects + interaction.
One-Sample Variance Test	A manufacturing company requires the diameter variance of machine-produced bolts to be $\leq 0.04 \text{ cm}^2$. A sample of bolts is used to test whether the variance exceeds this tolerance.
Two-Sample Variance Test	A bank checks whether the variance in monthly expenses differs between credit card users and debit card users , to understand financial stability differences.
One-Sample Proportion Test	A logistics company claims 95% of parcels are delivered on time. Out of 400 sampled deliveries, 365 were on time. Test if the on-time proportion is really 0.95.
Two-Sample Proportion Test	A political campaign tests if the approval rate for a candidate differs between urban voters and rural voters . Compares two independent proportions.

STATISTICAL TEST	REAL-LIFE EXAMPLE
Chi-Square Goodness-of-Fit	A zoo expects visitor choices of attractions to follow 50% mammals, 30% birds, 20% reptiles. Observed visitor counts differ, so they test whether the distribution fits the expected proportions.
Chi-Square Test of Association	A public health study tests whether smoking (yes/no) is associated with lung disease (yes/no) using a 2×2 contingency table.
One-Sample Poisson Test	A city historically receives 15 emergency calls per hour . Last week, the call center observed 22 calls in an hour. Test if the call rate has increased.
Two-Sample Poisson Test	Two factories are compared on accident occurrence rate : factory A had 18 accidents in 90 days; factory B had 25 accidents in 90 days. Test if their rates differ.
Pearson Correlation (Point-Biserial when one variable is binary)	A finance company tests whether income level is negatively correlated with the probability of loan default . Income is continuous; default is binary (0/1). Tests if higher income is associated with lower default risk.

Key Considerations

- **Sample Size:** Z-tests require large samples ($n \geq 30$); t-tests work with smaller samples
- **Known vs Unknown Variance:** Use z-tests when population SD is known, t-tests otherwise
- **Independence:** Independent samples vs paired/matched samples determine test choice
- **Assumptions:** Most parametric tests assume normality; check assumptions before applying
- **Multiple Groups:** Use ANOVA instead of multiple t-tests to control Type I error